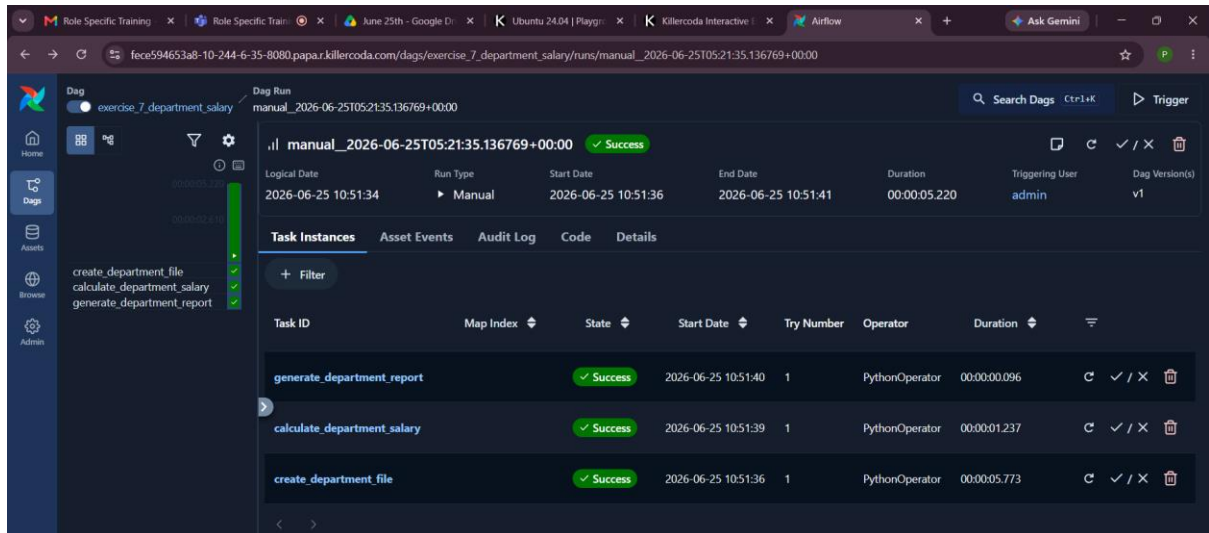


Capstone Project - Apache Airflow Workflows

Exercise 7: Department Salary Report

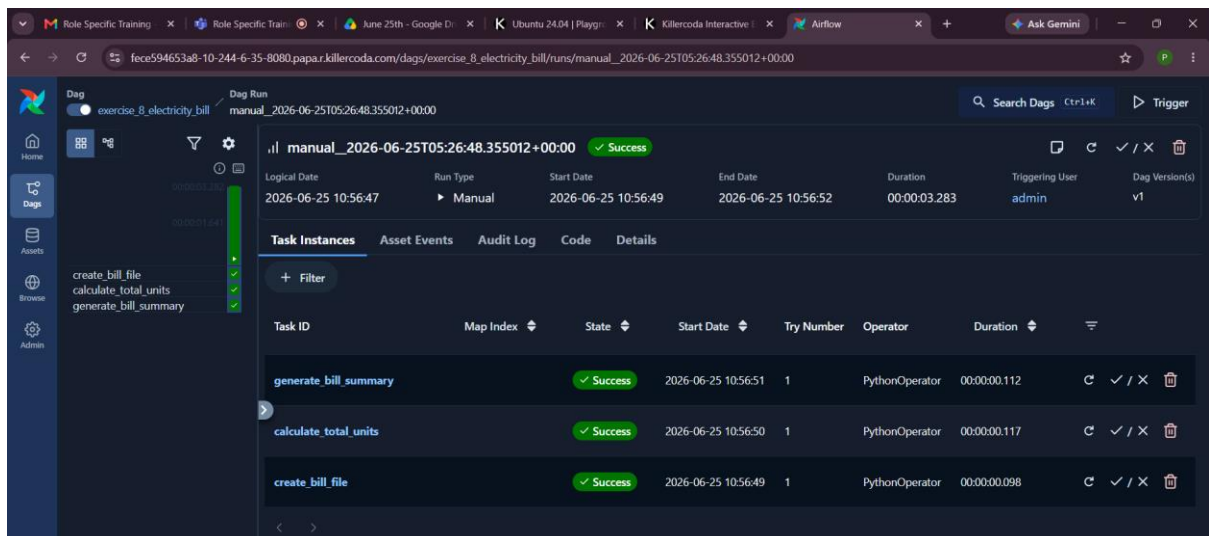


The screenshot shows the Apache Airflow web interface for the 'exercise_7_department_salary' DAG. The DAG run is 'manual_2026-06-25T05:21:35.136769+00:00' and it is in a 'Success' state. The logical date is '2026-06-25 10:51:34', the start date is '2026-06-25 10:51:36', the end date is '2026-06-25 10:51:41', and the duration is '00:00:05.220'. The triggering user is 'admin' and the DAG version is 'v1'. The task instances table shows three tasks: 'generate_department_report', 'calculate_department_salary', and 'create_department_file', all of which completed successfully.

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
generate_department_report		Success	2026-06-25 10:51:40	1	PythonOperator	00:00:00.096
calculate_department_salary		Success	2026-06-25 10:51:39	1	PythonOperator	00:00:01.237
create_department_file		Success	2026-06-25 10:51:36	1	PythonOperator	00:00:05.773

```
airflow@ba1f859f4a7a:/opt/airflow/dags$ cat /tmp/department_report.txt
IT=100000
HR=65000
Finance=90000
airflow@ba1f859f4a7a:/opt/airflow/dags$
```

Exercise 8: Electricity Bill Summary



The screenshot shows the Apache Airflow web interface for the 'exercise_8_electricity_bill' DAG. The DAG run is 'manual_2026-06-25T05:26:48.355012+00:00' and it is in a 'Success' state. The logical date is '2026-06-25 10:56:47', the start date is '2026-06-25 10:56:49', the end date is '2026-06-25 10:56:52', and the duration is '00:00:03.283'. The triggering user is 'admin' and the DAG version is 'v1'. The task instances table shows three tasks: 'generate_bill_summary', 'calculate_total_units', and 'create_bill_file', all of which completed successfully.

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
generate_bill_summary		Success	2026-06-25 10:56:51	1	PythonOperator	00:00:00.112
calculate_total_units		Success	2026-06-25 10:56:50	1	PythonOperator	00:00:00.117
create_bill_file		Success	2026-06-25 10:56:49	1	PythonOperator	00:00:00.098

```
Editor  Tab 1  +

airflow@ba1f859f4a7a:/opt/airflow/dags$ cat /tmp/bill_summary.txt
Customers = 5
Total Units = 1100
Average Units = 220
airflow@ba1f859f4a7a:/opt/airflow/dags$
```

Exercise 9: Exam Result Report

The screenshot shows the Apache Airflow web interface. The top navigation bar includes a search bar and a 'Trigger' button. The left sidebar contains icons for Home, Dags, Assets, Browse, Admin, Docs, and a clock icon. The main content area displays the details for a DAG run named 'manual_2026-06-25T05:29:09.093213+00:00'. The run status is 'Success'. Below this, there is a table of task instances.

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration	Actions
generate_result_summary		Success	2026-06-25 10:59:12	1	PythonOperator	00:00:00.097	✓ / ✕ 🗑️
count_pass_fail		Success	2026-06-25 10:59:11	1	PythonOperator	00:00:00.114	✓ / ✕ 🗑️
create_result_file		Success	2026-06-25 10:59:10	1	PythonOperator	00:00:00.093	✓ / ✕ 🗑️

```
Editor  Tab 1  +

airflow@ba1f859f4a7a:/opt/airflow/dags$ cat /tmp/result_summary.txt
Total Pass = 4
Total Fail = 2
airflow@ba1f859f4a7a:/opt/airflow/dags$
```

Exercise 10: Online Orders

```
Editor  Tab 1  +

airflow@ba1f859f4a7a:/opt/airflow/dags$ cat /tmp/sales_report.txt
Total Revenue = 100500
Highest Selling Product = Laptop
airflow@ba1f859f4a7a:/opt/airflow/dags$
```

Role Specific Training x Role Specific Train x June 25th - Google E x K Ubuntu 24.04 | Playg x K Killercode Interactive x Airflow x Ask Gemini

fece594653a8-10-244-6-35-8080.papar.killercode.com/dags/exercise_10_online_orders/runs/manual_2026-06-25T05:32:00.582029+00:00

Dag Run manual_2026-06-25T05:32:00.582029+00:00

Search Dags Ctrl+K Trigger

exercise_10_online_orders

Logical Date 2026-06-25 11:01:58 Run Type Manual Start Date 2026-06-25 11:02:01 End Date 2026-06-25 11:02:04 Duration 00:00:03.308 Triggering User admin Dag Version(s) v1

Task Instances Asset Events Audit Log Code Details

+ Filter

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration	
generate_sales_report		Success	2026-06-25 11:02:03	1	PythonOperator	00:00:01.116	✓ / ✕
calculate_order_value		Success	2026-06-25 11:02:02	1	PythonOperator	00:00:00.107	✓ / ✕
create_orders		Success	2026-06-25 11:02:01	1	PythonOperator	00:00:00.108	✓ / ✕

create_orders
calculate_order_value
generate_sales_report

Docs +05:30 User

Exercise 11: Temperature Analysis

Role Specific Training x Role Specific Train x June 25th - Google E x K Ubuntu 24.04 | Playg x K Killercode Interactive x Airflow x Ask Gemini

c93e5c767e29-10-244-5-182-8080.papar.killercode.com/dags/exercise_11_temperature_analysis/runs/manual_2026-06-25T09:36:53.124726+00:00

Dag Run manual_2026-06-25T09:36:53.124726+00:00

Search Dags Ctrl+K Trigger

exercise_11_temperature_analysis

Logical Date 2026-06-25 15:06:53 Run Type Manual Start Date 2026-06-25 15:06:54 End Date 2026-06-25 15:07:01 Duration 00:00:06.983 Triggering User admin Dag Version(s) v1

Task Instances Asset Events Audit Log Code Details

+ Filter

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration	
generate_weather_report		Success	2026-06-25 15:07:00	1	PythonOperator	00:00:00.204	✓ / ✕
find_highest_temperature		Success	2026-06-25 15:06:59	1	PythonOperator	00:00:00.231	✓ / ✕
create_temperature_file		Success	2026-06-25 15:06:54	1	PythonOperator	00:00:07.502	✓ / ✕

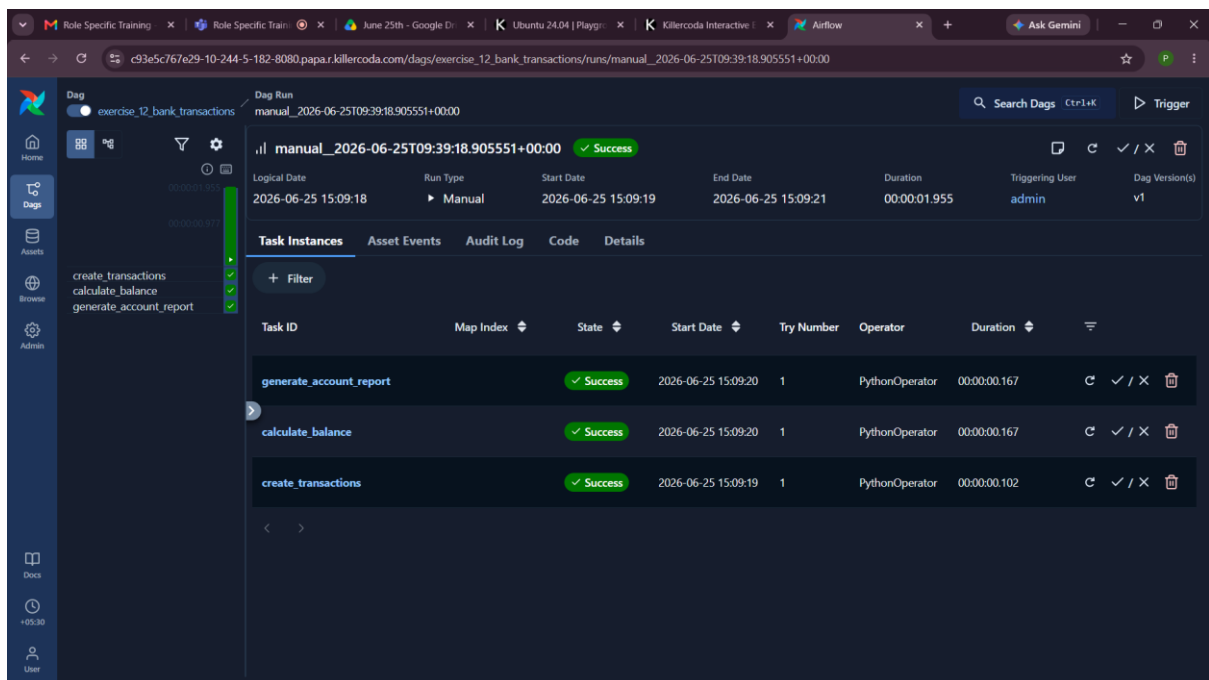
create_temperature_file
find_highest_temperature
generate_weather_report

Docs +05:30 User

Editor Tab 1 +

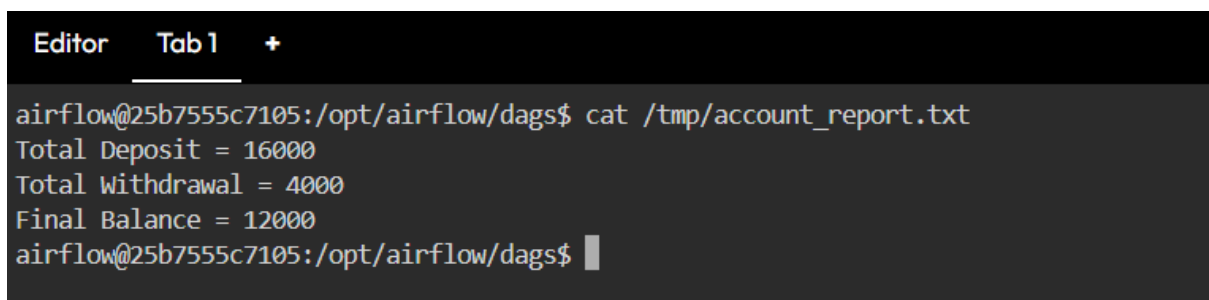
```
airflow@25b7555c7105:/opt/airflow/dags$ cat /tmp/weather_report.txt
Highest Temperature = 38
Average Temperature = 34.14
airflow@25b7555c7105:/opt/airflow/dags$
```

Exercise 12: Bank Transaction Summary



The screenshot displays the Apache Airflow web interface. On the left, a sidebar shows navigation options: Home, Dags, Assets, Browse, and Admin. The 'Dags' section is active, showing a list of DAGs including 'create_transactions', 'calculate_balance', and 'generate_account_report'. The main panel shows the details of a specific DAG run: 'manual_2026-06-25T09:39:18.905551+00:00'. The run is marked as 'Success'. Below this, the 'Task Instances' tab is selected, showing a table of task execution details.

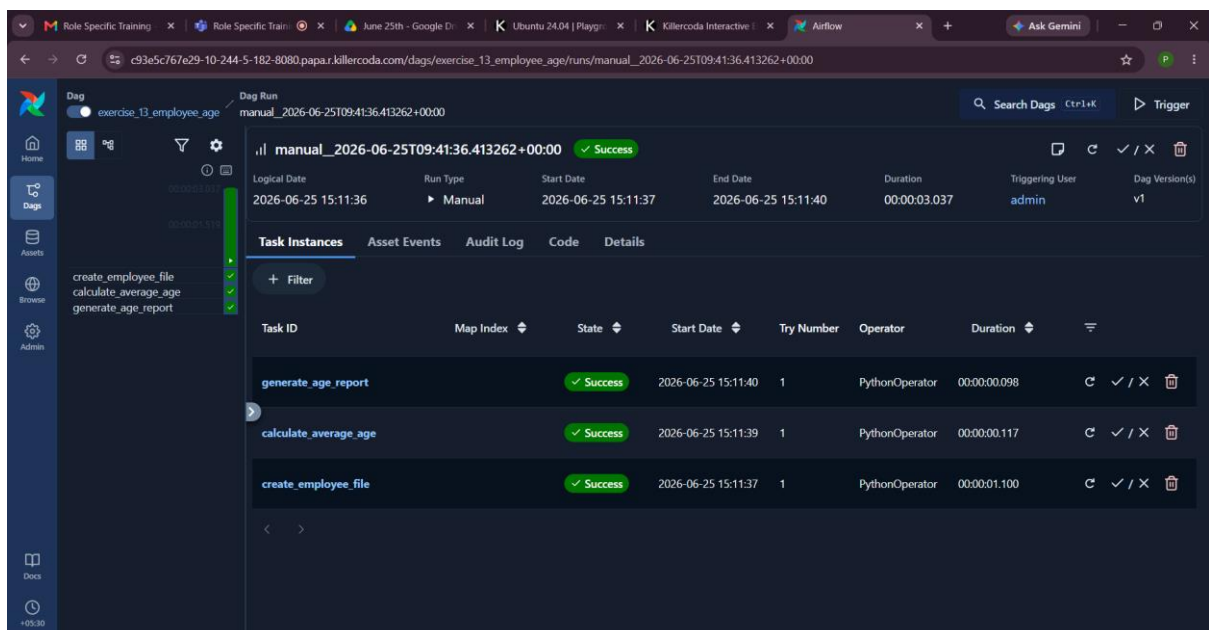
Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
generate_account_report		Success	2026-06-25 15:09:20	1	PythonOperator	00:00:00.167
calculate_balance		Success	2026-06-25 15:09:20	1	PythonOperator	00:00:00.167
create_transactions		Success	2026-06-25 15:09:19	1	PythonOperator	00:00:00.102



The screenshot shows a terminal window with the following text:

```
airflow@25b7555c7105:/opt/airflow/dags$ cat /tmp/account_report.txt
Total Deposit = 16000
Total Withdrawal = 4000
Final Balance = 12000
airflow@25b7555c7105:/opt/airflow/dags$
```

Exercise 13: Employee Age Report



The screenshot displays the Apache Airflow web interface. On the left, a sidebar shows navigation options: Home, Dags, Assets, Browse, and Admin. The 'Dags' section is active, showing a list of DAGs including 'create_employee_file', 'calculate_average_age', and 'generate_age_report'. The main panel shows the details of a specific DAG run: 'manual_2026-06-25T09:41:36.413262+00:00'. The run is marked as 'Success'. Below this, the 'Task Instances' tab is selected, showing a table of task execution details.

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
generate_age_report		Success	2026-06-25 15:11:40	1	PythonOperator	00:00:00.098
calculate_average_age		Success	2026-06-25 15:11:39	1	PythonOperator	00:00:00.117
create_employee_file		Success	2026-06-25 15:11:37	1	PythonOperator	00:00:01.100

```
Editor  Tab 1  +

airflow@25b7555c7105:/opt/airflow/dags$ cat /tmp/age_report.txt
Youngest Employee = Sneha
Oldest Employee = Amit
Average Age = 33.0
airflow@25b7555c7105:/opt/airflow/dags$
```

Exercise 14: Course Enrollment Report

The screenshot shows the Apache Airflow web interface. The top navigation bar includes links for Home, Dags, Assets, Browse, Admin, Docs, and a clock showing 10:53. The main content area displays the details for a DAG run named 'manual_2026-06-25T09:43:58.861843+00:00' with a status of 'Success'. Below this, the 'Task Instances' tab is active, showing a table of task execution details.

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration	Actions
generate_course_report		Success	2026-06-25 15:14:01	1	PythonOperator	00:00:00.095	Copy / Refresh / Delete
count_students		Success	2026-06-25 15:14:00	1	PythonOperator	00:00:00.106	Copy / Refresh / Delete
create_enrollment_file		Success	2026-06-25 15:13:59	1	PythonOperator	00:00:00.089	Copy / Refresh / Delete

```
Editor  Tab 1  +

airflow@25b7555c7105:/opt/airflow/dags$ cat /tmp/course_report.txt
Python = 3
SQL = 2
Power BI = 2
airflow@25b7555c7105:/opt/airflow/dags$
```